



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)

REVIEW TEST

06-06-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]

PAPER-2

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **29** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

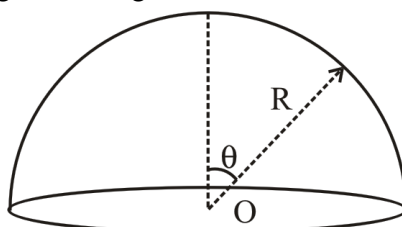
Please see the last page of this booklet for rest of the instructions

PHYSICS

SEC-1 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Select the correct option for the given configuration.



Hemispherical shell, having surface charge density σ (may be constant or variable)

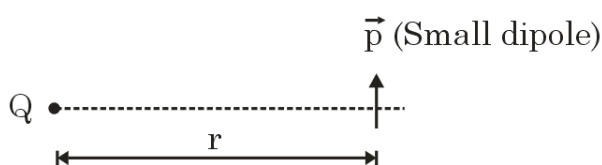
(A) Electric field at O (if $\sigma = \sigma_0 \cos\theta$) = $\frac{\sigma_0}{6\epsilon_0}$

(B) Electric field at O (if $\sigma = \sigma_0 = \text{constant}$) = $\frac{\sigma_0}{2\epsilon_0}$

(C) Potential at O due to charge on hemisphere if ($\sigma = \sigma_0 = \text{constant}$) = $\frac{\sigma_0 R}{2\epsilon_0}$

(D) Potential at O due to charge on hemisphere if ($\sigma = \sigma_0 \cos\theta$) = $\frac{\sigma_0 R}{2\epsilon_0}$

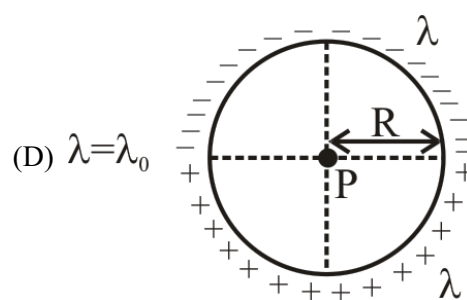
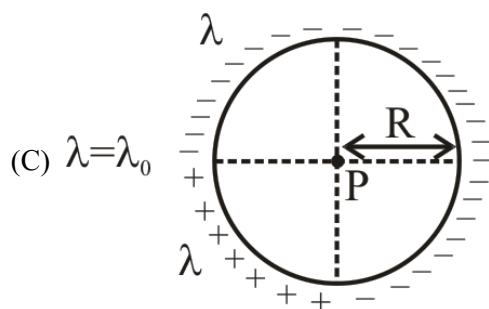
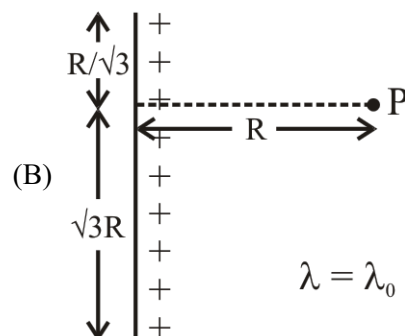
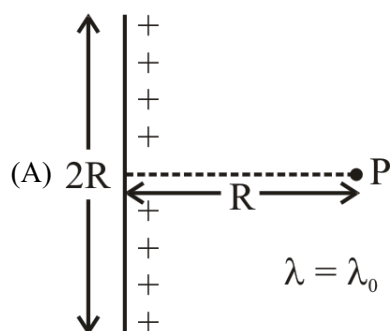
2. A point charge $+Q$ is fixed at a point O . Another point charge $+q$ & mass m is projected from infinity towards $+Q$ with speed v_0 . Minimum distance of approach is r . If it is now given speed $2v_0$ at infinity then
- (A) Minimum distance of approach becomes $\frac{r}{2}$
- (B) Minimum distance of approach becomes $\frac{r}{4}$
- (C) Angular momentum conservation will hold true about O .
- (D) Mechanical energy will remain conserved throughout the motion.
3. For the situation shown in the figure (assume $r \gg$ length of dipole) mark out the correct statement(s).



- (A) Force acting on the dipole is zero
- (B) Force acting on the dipole is approximately $\frac{pQ}{4\pi\epsilon_0 r^3}$ and is acting upward
- (C) Torque acting on the dipole is $\frac{pQ}{4\pi\epsilon_0 r^2}$ in clockwise direction
- (D) Torque acting on dipole is $\frac{pQ}{4\pi\epsilon_0 r^2}$ in anti-clockwise direction

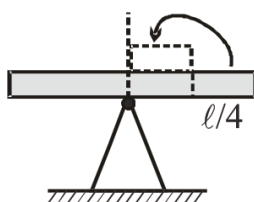
Space for Rough Work

4. For which of the following continuous charge system magnitude of electric field at point P is $\frac{\sqrt{2}k\lambda_0}{R}$ where λ_0 is linear charge density.



Space for Rough Work

5. Two point objects of masses m and $4m$ are at rest at an infinite separation. They move towards each other under mutual gravitational attraction. If G is the universal gravitational constant, then at a separation r .
- (A) the total mechanical energy of the two objects is zero
- (B) their relative velocity is $\sqrt{\frac{10Gm}{r}}$
- (C) the total kinetic energy of the objects is $\frac{4Gm^2}{r}$
- (D) their relative velocity is zero
6. A uniform rod of mass ' m ' and length ' ℓ ' is balanced on a triangular prism. (see figure). Now, one end of rod is cut by $\frac{\ell}{4}$ and it is placed on the shortened end of the rod. Then



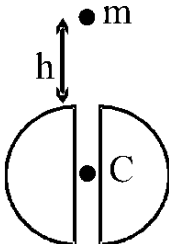
- (A) the rod will remain in equilibrium
- (B) the centre of mass of the rod will shift to the left.
- (C) the rod starts rotating anticlockwise.
- (D) the rod will start rotating clockwise.

Space for Rough Work

7. A gymnast throws the hoop of mass 'm' and radius 'R' horizontally on a rough horizontal surface to the right with speed ' v ' and angular velocity $\omega = 2v/R$ as shown in the figure :-



- (A) The angular velocity of hoop, when it stops translating is v/R
- (B) The distance traveled by the hoop before it stops translating is $\frac{v^2}{2\mu g}$
- (C) The time for which the hoop slips on ground is $\frac{3v}{2\mu g}$
- (D) Work done by friction during the process of motion is $\frac{7mv^2}{4}$
8. A ball of mass m is dropped from a height h equal to the radius of the Earth (R) above the tunnel dug through the Earth as shown in the figure. Ignore the effect of earth's rotation and air resistance, M is mass of Earth. Choose the correct alternative(s) :-



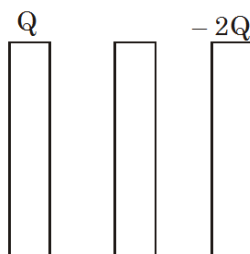
- (A) ball will oscillate through the Earth to a height h on both sides
- (B) ball will execute simple harmonic motion.
- (C) motion of the ball is periodic
- (D) ball passes the centre of Earth with a speed $v = \sqrt{\frac{2GM}{R}}$

Space for Rough Work

SEC-2 : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

1. Match the entries on left side with entries on right side. Three identical metal plates with large surface areas are kept parallel to each other as shown in figure. The left most is given a charge Q , the right most a charge $-2Q$ and the middle one remains neutral. Then

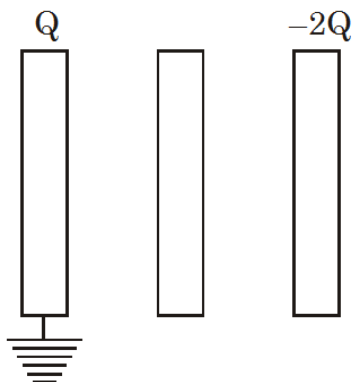


List-I		List-II	
(I)	The charge appearing on outer surface of right most plate is	(P)	$+\frac{Q}{2}$
(II)	The charge appearing on outer surface of left most plate is	(Q)	$-\frac{Q}{2}$
(III)	The charge appearing on left surface of middle plate is	(R)	$-\frac{3Q}{2}$
(IV)	The charge appearing on right surface of middle plate is	(S)	$\frac{3Q}{2}$
		(T)	$+Q$
		(U)	$-Q$

- (A) $I \rightarrow P; II \rightarrow Q; III \rightarrow R; IV \rightarrow U$ (B) $I \rightarrow P; II \rightarrow Q; III \rightarrow R; IV \rightarrow S$
 (C) $I \rightarrow P; II \rightarrow T; III \rightarrow R; IV \rightarrow U$ (D) $I \rightarrow Q; II \rightarrow Q; III \rightarrow R; IV \rightarrow S$

Space for Rough Work

2. Match the entries on left side with entries on right side when left most plate is grounded. Then

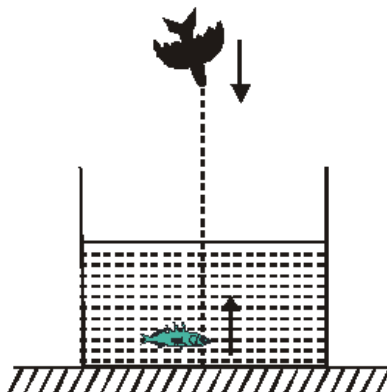


List-I		List-II	
(I)	The charge appearing on outer surface of right most plate is	(P)	0
(II)	The charge appearing on outer surface of left most plate is	(Q)	Q
(III)	The charge appearing on left surface of middle plate is	(R)	$-Q$
(IV)	The charge appearing on right surface of middle plate is	(S)	$2Q$
		(T)	$-2Q$
		(U)	$3Q$

- (A) $I \rightarrow P; II \rightarrow Q; III \rightarrow R; IV \rightarrow U$ (B) $I \rightarrow P; II \rightarrow P; III \rightarrow T; IV \rightarrow U$
 (C) $I \rightarrow P; II \rightarrow P; III \rightarrow T; IV \rightarrow S$ (D) $I \rightarrow Q; II \rightarrow Q; III \rightarrow R; IV \rightarrow S$

Space for Rough Work

3. A bird in air is diving vertically over a tank with speed 6 cm/s. Base of the tank is silvered. A fish in the tank is rising upward along the same line with speed 4 cm/s. [Take: $\mu_{\text{water}} = 4/3$]



List-I		List-II	
(I)	Speed (in cm/s) of the image of fish as seen by the bird directly.	(P)	12
(II)	Speed (in cm/s) of the image of fish formed after reflection from the mirror as seen by the bird.	(Q)	4
(III)	Speed (in cm/s) of image of bird relative to the fish looking upwards.	(R)	9
(IV)	Speed (in cm/s) of image of bird relative to the fish looking downwards in the mirror	(S)	3
		(T)	20/3
		(U)	32/3

(A) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ Q

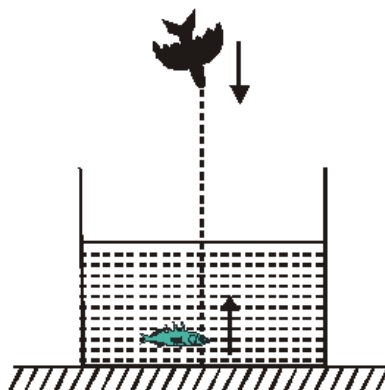
(B) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ P; IV $\rightarrow$ T

(C) I $\rightarrow$ R; II $\rightarrow$ S; III $\rightarrow$ U; IV $\rightarrow$ Q

(D) I $\rightarrow$ R; II $\rightarrow$ T; III $\rightarrow$ P; IV $\rightarrow$ Q

Space for Rough Work

4. A bird in air is diving vertically over a tank with speed 6 cm/s. Base of the tank is silvered. A fish in the tank is rising upward along the same line with speed 4 cm/s. Water level is falling at 4 cm/s. [Take: $\mu_{\text{water}} = 4/3$]



List-I		List-II	
(I)	Speed (in cm/s) of the image of fish as seen by the bird directly.	(P)	8
(II)	Speed (in cm/s) of the image of fish formed after reflection from the mirror as seen by the bird.	(Q)	2
(III)	Speed (in cm/s) of image of bird relative to the fish looking upwards.	(R)	$8/3$
(IV)	Speed (in cm/s) of image of bird relative to the fish looking downwards in the mirror.	(S)	3
		(T)	$20/3$
		(U)	$32/3$

- (A) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ U; IV $\rightarrow$ T
 (B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ T; IV $\rightarrow$ R
 (C) I $\rightarrow$ P; II $\rightarrow$ U; III $\rightarrow$ U; IV $\rightarrow$ R
 (D) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ U; IV $\rightarrow$ R

Space for Rough Work

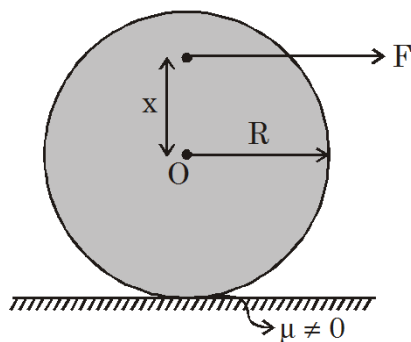
SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

1. There are 8 drops of a conducting fluid. Each has radius r and they are charged to potential 1 volt. They are then combined to form a bigger drop. Find potential of big drop in volt.
2. A simple pendulum has a time period T_1 when on earth's surface, and T_2 when taken at a height $2R$ above the earth's surface where R is the radius of earth. The value of T_2/T_1 is :-
3. A system of binary stars of masses M_1 and M_2 are moving in circular orbits of radii r_1 and r_2 respectively. If T_1 and T_2 are the time periods of revolution of masses M_1 and M_2 respectively and if $\frac{r_1}{r_2} = 2$, then find $\frac{T_1}{T_2}$.
4. A uniform disk of mass m and radius R rolls, without slipping, down a fixed plane inclined at an angle 30° to the horizontal. The linear acceleration of the disk (in m/sec^2) is

Space for Rough Work

5. Initially a disc of radius 30 cm was in rest & we apply force F as shown in figure. If friction acting on disc is zero then find the value of x (in cm).



6. Two point charges A and B of charge $+4\mu\text{C}$ and $-1\mu\text{C}$ are placed in air at a distance of one metre apart. The distance (in m) of the point on the line joining the charges and from the charge B, where the resultant electric field is zero, will be :

Space for Rough Work

CHEMISTRY

SEC-1 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

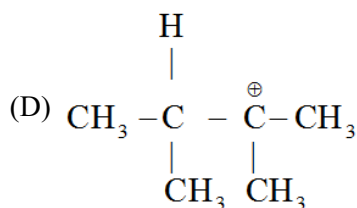
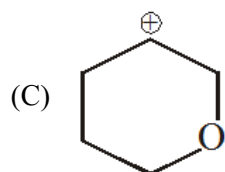
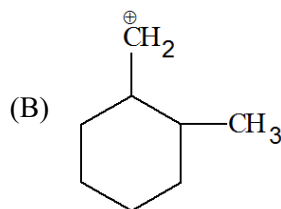
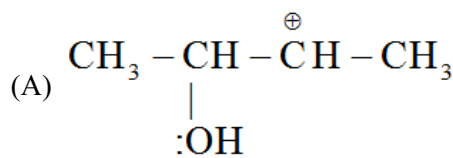
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

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- For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Which of the following is/are shows re-arrangement ?



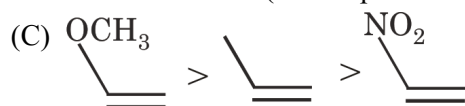
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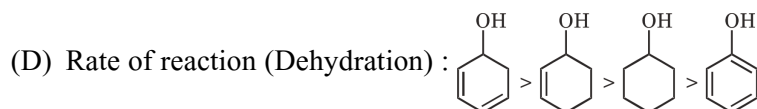
2. Which order is correct :

(A) Nucleophilicity (In DMSO) : $\text{F}^\ominus > \text{Cl}^\ominus > \text{Br}^\ominus > \text{I}^\ominus$

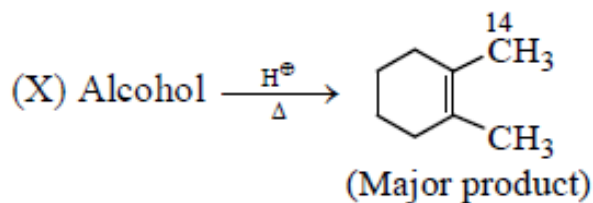
(B) Leaving group tendency : $\text{N}_2 > \text{O}^\ominus - \overset{\text{O}}{\parallel} \text{C} - \text{CH}_3 > \text{OH}^\ominus$

Rate of reaction (Electrophillic addition reaction) :

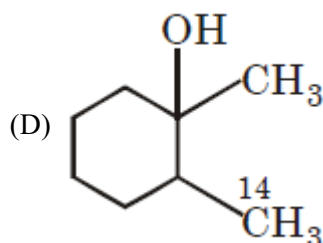
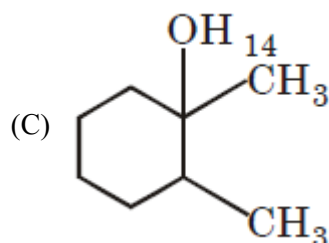
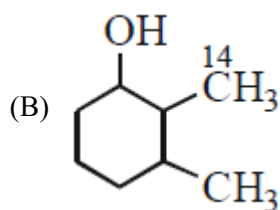
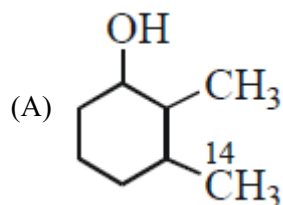
(C) 

(D) Rate of reaction (Dehydration) : 

3.



(X) may be

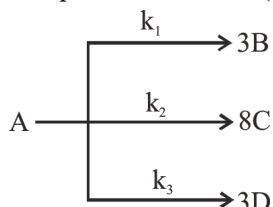


Space for Rough Work

4. Select the correct statement(s)

- (A) System having non-permeable and adiabatic boundary must be isolated
- (B) Molarity, normality & molality all are intensive properties
- (C) $\Delta U = q + w$ can be used for any system.
- (D) Closed system can transfer energy only.

5. For parallel reaction (all following first order kinetics)



at time $t = 0$, initial mole of A is 1. If overall half life of reaction is 15 days and $k_1 : k_2 : k_3 = 4 : 2 : 1$, then $[\log 2 = 0.3, 2^{1/3} = 5/4]$

- (A) Number of moles of C after 45 days are 2.
- (B) Number of moles of B after 30 days are $\frac{36}{28}$.
- (C) Total moles at the end of reaction are 3.
- (D) Ratio of moles of B & D at any time (except $t = 0$) is 4.

6. Select the correct statement(s)

- (A) Liquid solution of bromoethane & chloroethane obey's Raoult's law
- (B) Liquid solution of EtOH & $\text{H}_3\text{C-COCH}_3$ exerts more pressure, than predicted by Raoult's law
- (C) Mixture of CHCl_3 and H_3COCH_3 exerts pressure less than predicted by Raoult's law
- (D) On mixing benzene & toluene, ΔH_{mix} & ΔV_{mix} both are positive.

Space for Rough Work

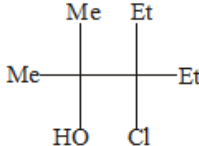
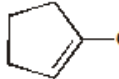
7. The **CORRECT** statement is / are :
- (A) BeF_2 is more soluble in water than BaF_2
 - (B) AgF is colourless while AgI is coloured, due to higher polarisability of F^-
 - (C) BiF_5 exists but BiI_5 does not exist
 - (D) BeCO_3 is less thermally stable than BaCO_3
8. Select the **CORRECT** order of radii :-
- (A) $\text{Na}^+ > \text{Al}^{3+}$
 - (B) $\text{N}^{3-} > \text{F}^-$
 - (C) $\text{Y} > \text{La}$
 - (D) $\text{Sc} > \text{Mn}$

Space for Rough Work

SEC-2 : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- Four** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

1.

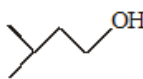
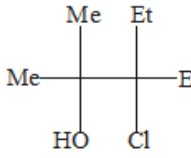
	List-I		List-II
(I)	 $\xrightarrow{H^+ / \Delta}$ Major product	(P)	Major product is Saytzeff's alkene
(II)	$Ph-CH_2-CH=CH_2 \xrightarrow{HBr}$ Major product	(Q)	Major product is Hoffmann's alkene
(III)	 $\xrightarrow{AlCl_3 / \Delta}$ Major product	(R)	Rearrangement takes place
(IV)	 $\xrightarrow{HCl \text{ in } H_2O}$ Major product	(S)	Markonikov addition takes place
		(T)	Carbocation is form as intermediate
		(U)	Alcohol is form as a major product

Which of the following option has correct combination considering List-I and List-II

- (A) IV $\rightarrow$ S,T,U (B) III $\rightarrow$ Q,U (C) II $\rightarrow$ Q, R (D) I $\rightarrow$ Q,U

Space for Rough Work

2.

	List-I		List-II
(I)	 $\xrightarrow{H^+/\Delta}$ Major product	(P)	Major product is Saytzeff's alkene
(II)	$Ph-CH_2-CH=CH_2 \xrightarrow{HBr}$ Major product	(Q)	Major product is Hoffmann's alkene
(III)	 $\xrightarrow{AlCl_3/\Delta}$ Major product	(R)	Rearrangement takes place
(IV)	 $\xrightarrow{HCl \text{ in } H_2O}$ Major product	(S)	Markonikov addition takes place
		(T)	Carbocation is form as intermediate
		(U)	Alcohol is form as a major product

Choose the incorrect combination considering List-I and List-II

- (A) I $\rightarrow$ P,R,T (B) II $\rightarrow$ R,S,T
(C) III $\rightarrow$ Q, U (D) IV $\rightarrow$ S, T, U

3.

	List-I Complex		List-II Characteristics given complexes
(I)	$[Co(en)_2(NO_2)_2]Cl$	(P)	Maximum number of N–Co–N bond angles
(II)	$Na[Co(EDTA)]$	(Q)	Maximum number of O–Co–O bond angles
(III)	$Hg[Co(SCN)_4]$	(R)	Have ambident ligand(s)
(IV)	$[Co(Gly)_3]^0$	(S)	Maximum number of five member chelate rings
		(T)	Oxidation number of metal is +2
		(U)	C.N. of metal is 3

Select **INCORRECT** combination for list-I and list-II

- (A) I $\rightarrow$ P (B) II $\rightarrow$ Q (C) III $\rightarrow$ R (D) IV $\rightarrow$ T

Space for Rough Work

4.

	List-I Complex		List-II Characteristics given complexes
(I)	$[\text{Co(en)}_2(\text{NO}_2)_2]\text{Cl}$	(P)	Maximum number of N–Co–N bond angles
(II)	$\text{Na}[\text{Co(EDTA)}]$	(Q)	Maximum number of O–Co–O bond angles
(III)	$\text{Hg}[\text{Co(SCN)}_4]$	(R)	Have ambidentated ligend(s)
(IV)	$[\text{Co(Gly)}_3]^0$	(S)	Maximum number of five member chelate rings
		(T)	Oxidation number of metal is +2
		(U)	C.N. of metal is 3

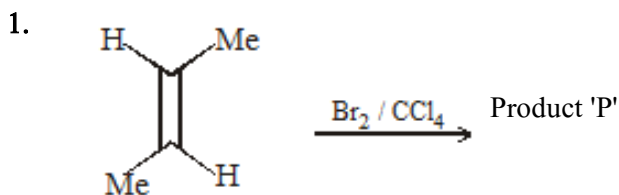
 Select **CORRECT** combination for list-I and list-II

- (A) I $\rightarrow$ S
 (B) II $\rightarrow$ T
 (C) III $\rightarrow$ T
 (D) IV $\rightarrow$ S

 Space for Rough Work

SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.



How many stereo isomer of product "P" is/are possible?

- A sample of $^{131}_{53}\text{I}$, as I^- ion, was administered to a patient in a carrier consisting 1.0 mg of stable I^- ion. After 4.0 days, 60% of the initial radioactivity was detected in the thyroid gland of the patient. What mass (in mg) of the stable I^- ion had migrated to the thyroid gland? Given $t_{1/2}$ to $\text{I}^{131} = 8$ days
- 20 gm of a non volatile solute B when added to 500 gm of a volatile solvent A, decreases the vapour pressure of pure solvent A from 306 torr to 300 torr at 300 K. What is the molecular mass of non-volatile solute (g/mole). The molecular mass of solvent is 25 g/mole.
- At 373 K, a gaseous reaction $\text{A(g)} \rightarrow 2\text{B(g)} + \text{C(g)}$ is found to be of first order. Starting with pure A, the total pressure at the end of 10 min was 176 mm Hg and after a long time when A was completely dissociated, it was 270 mm Hg. Then pressure of A (in mm Hg) at the end of 10 minutes was :
- How many compound(s) gives diprotic acid on hydrolysis ?
 SO_2Cl_2 , SOCl_2 , POCl_3 , XeF_2 , N_2O_5 , P_4O_6 , SF_4 , BCl_3
- Number of species which have higher " C-O " bond length as compare to carbon monoxide (CO)
 CO_2 , CO_3^{2-} , CH_3O^- , HCOO^- , CH_3COO^- , $[\text{V}(\text{CO})_6]^+$, $[\text{Cr}(\text{CO})_6]$, $[\text{Fe}(\text{CO})_4]^{2-}$

Space for Rough Work

MATHEMATICS

SEC-1 : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

1. Which of the following is/are correct ?

(A) $(\tan x)^{\ell_n(\sin x)} > (\cot x)^{\ell_n(\sin x)}, \forall x \in (0, \pi/4)$

(B) $4^{\ell_{ncosec x}} < 5^{\ell_{ncosec x}}, \forall x \in (0, \pi/2)$

(C) $\left(\frac{1}{e}\right)^{\ell_n(\cos x)} < \left(\frac{1}{3}\right)^{\ell_n(\cos x)}, \forall x \in (0, \pi/2)$

(D) $2^{\log_{1/\pi}(\tan x)} < 2^{\log_{1/\pi}(\sin x)}, \forall x \in (0, \pi/2)$

Space for Rough Work

2. Consider the equation $2 \cot^2 \theta + 2\sqrt{3} \cot \theta + 4 \operatorname{cosec} \theta + 8 = 0$
- (A) The general solution of the equation is $2n\pi - \frac{\pi}{6}, n \in \mathbb{I}$
- (B) The general solution of the equation is $2n\pi + \frac{\pi}{6}, n \in \mathbb{I}$
- (C) Number of principal solutions of the equation is 4.
- (D) Number of principal solutions of the equation is 1.
3. If $y = xe^{x+xe^{x+xe^{x+\dots\infty}}}$, $x, y \in \mathbb{R}^+$, then -
- (A) $\log x - \log y = x + y$ (B) $\log x - \log y = x - y$
- (C) $\frac{dy}{dx} = \frac{(x+1)y}{(1-y)x}$ (wherever exists) (D) $\frac{dy}{dx} = \frac{(x-1)y}{(1+y)x}$
4. Let $f(x) = \begin{cases} x^7 + 2x^3 - 3x^2 + 6x + 101; & x \leq 1 \\ 107x^2 + \sin(1-x) & ; x > 1 \end{cases}$, then identify the CORRECT statement ?
- (A) $f(x)$ is decreasing in $\mathbb{R}$
- (B) $f\left(\frac{1}{101}\right) > f\left(\frac{1}{107}\right)$
- (C) $f(107+x) > f(107-x) \forall x \in \mathbb{R}$
- (D) $f(x+101) > f(x-101) \forall x \in \mathbb{R}$

Space for Rough Work

5. Let $f, g : [-2, 3] \rightarrow \mathbb{R}$ be continuous function which are twice differentiable on the interval $(-2, 3)$. Let the values of f and g at the points $-2, 0, 1$ and 3 be as given in the following table :

	$x = -2$	$x = 0$	$x = 1$	$x = 3$
$f(x)$	11	-4	6	16
$g(x)$	1	-2	0	2

In each of the intervals $(-2, 0)$, $(0, 1)$ and $(1, 3)$ the function $(f - 5g)''$ never vanishes. Then the correct statement(s) is (are)

- (A) $f'(x) - 5g'(x) = 0$ has exactly three solutions in $(-2, 0) \cup (0, 1) \cup (1, 3)$
- (B) $f'(x) - 5g'(x) = 0$ has exactly one solutions in $(0, 1)$
- (C) $f'(x) - 5g'(x) = 0$ has at least one solution in $(1, 3)$
- (D) $f'(x) - 5g'(x) = 0$ has at least two solution in $(-2, 0)$
6. Let $f(x) = \begin{cases} 3 \sin x + a^2 - 10a + 30 & \text{if } x \notin \mathbb{Q} \\ 4 \cos x & \text{if } x \in \mathbb{Q} \end{cases}$, then which of the following statement(s) is/are false ?

- (A) $f(x)$ is continuous for all x when $a = 5$.
- (B) $f(x)$ is discontinuous for all x when $a = 5$.
- (C) $f(x)$ is continuous for all $x = 2n\pi - \tan^{-1}\left(\frac{3}{4}\right), n \in \mathbb{I}$ when $a = 5$.
- (D) $f(x)$ is continuous for all $x = 2n\pi - \tan^{-1}\left(\frac{4}{3}\right), n \in \mathbb{I}$ when $a = 5$.

Space for Rough Work

7. Let $y = f(x)$ be a continuous and twice differentiable function such that $f(1) = f(3) = f(5) = 0$, then in $(1, 5)$ equation
- (A) $2f(x) = f'(x)$ has at least 2 real roots (B) $2f(x) = f'(x)$ has exactly two real roots
- (C) $f''(x) - 4f'(x) + 4f(x) = 0$ has a real root (D) $f''(x) = 0$ has a real root
8. For the function $f(x) = x^2 + x + 1 + \sin x$
- (A) $f''(x)$ is monotonically increasing
- (B) $f'(x)$ is monotonically increasing
- (C) $f'(x) = 0$ has exactly one root in $(-1, 0)$
- (D) $f'(x) = 0$ has atleast two distinct root in $(-1, 0)$

Space for Rough Work

SEC-2 : (Maximum Marks: 12)

- This section contains **TWO (02)** List-Match sets.
- Section contains **TWO** List-Match sets.
- Each List-Match set has **Two** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Six** entries (P), (Q), (R), (S), (T) and (U)
- **Four** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks : +3 If **ONLY** the option corresponding to the correct combination is chosen.
Zero Marks : 0 If none of the options is chosen (i.e., the question is unanswered);
Negative Marks : -1 In all other cases

1. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(I)	If angle of intersection of curves $x^2 + y^2 - 4x + 6y + 4 = 0$ and $2x - y - 7 = 0$ is $\frac{\pi}{k}$, then 'k' can be	(P)	3
(II)	A rectangle is inscribed in an equilateral triangle of side length 4. If the maximum area of rectangle is p, then [p] equals (Where [.] denotes greatest integer function)	(Q)	4
(III)	Let $f(x) = \tan^{-1} \sqrt{\frac{1+\sin x}{1-\sin x}}$ for $x \in \left(\frac{\pi}{2}, \pi\right)$, then $\left \frac{2}{f'(\frac{2\pi}{3})} \right $ is equal to	(R)	1
(IV)	Let $y = 2 \tan^{-1} x$ and $z = \sin^{-1} \frac{2x}{1+x^2}$, then $\left \frac{dy}{dz} \right $ equals	(S)	2
		(T)	5
		(U)	6

Which of the following is the only **CORRECT** combination ?

- (A) $I \rightarrow R$ (B) $I \rightarrow Q$ (C) $II \rightarrow P$ (D) $II \rightarrow Q$

Space for Rough Work

2. Match List-I with List-II and select the correct answer using the code given below the list.

List-I		List-II	
(I)	If angle of intersection of curves $x^2 + y^2 - 4x + 6y + 4 = 0$ and $2x - y - 7 = 0$ is $\frac{\pi}{k}$, then 'k' can be	(P)	3
(II)	A rectangle is inscribed in an equilateral triangle of side length 4. If the maximum area of rectangle is p, then [p] equals (Where [.] denotes greatest integer function)	(Q)	4
(III)	Let $f(x) = \tan^{-1} \sqrt{\frac{1+\sin x}{1-\sin x}}$ for $x \in \left(\frac{\pi}{2}, \pi\right)$, then $\left \frac{2}{f'(\frac{2\pi}{3})} \right $ is equal to	(R)	1
(IV)	Let $y = 2 \tan^{-1} x$ and $z = \sin^{-1} \frac{2x}{1+x^2}$, then $\left \frac{dy}{dz} \right $ equals	(S)	2
		(T)	5
		(U)	6

Which of the following is the only **CORRECT** combination ?

- (A) III $\rightarrow$ Q (B) III $\rightarrow$ R (C) IV $\rightarrow$ S (D) IV $\rightarrow$ P

Space for Rough Work

3. Let $f(x) = x^3 - 6x^2 - 36x + 2$

List-I		List-II	
(I)	$f(x)$ is increasing function in	(P)	$\left(0, \frac{3}{2}\right)$
(II)	The tangent to $y = f(x)$ at $x = 0$, meet the curve again at point P, whose abscissae lies in	(Q)	$\left(\frac{3}{2}, 4\right)$
(III)	Let N denotes number of real root(s) of equation $f(x) = 0$, then N lies in	(R)	(5,7)
(IV)	Let Rolle's theorem holds for $f(x)$ in $[0, a]$. If k is sum of all possible values of a, then k lies in	(S)	(8,10)
		(T)	(9,11)
		(U)	$\left(\frac{1}{2}, \frac{1}{4}\right)$

Which of the following is the only **INCORRECT** combination ?

- (A) $I \rightarrow S$ (B) $I \rightarrow T$ (C) $III \rightarrow P$ (D) $III \rightarrow Q$

Space for Rough Work

4. Let $f(x) = x^3 - 6x^2 - 36x + 2$

List-I		List-II	
(I)	$f(x)$ is increasing function in	(P)	$\left(0, \frac{3}{2}\right)$
(II)	The tangent to $y = f(x)$ at $x = 0$, meet the curve again at point P, whose abscissae lies in	(Q)	$\left(\frac{3}{2}, 4\right)$
(III)	Let N denotes number of real root(s) of equation $f(x) = 0$, then N lies in	(R)	(5,7)
(IV)	Let Rolle's theorem holds for $f(x)$ in $[0, a]$. If k is sum of all possible values of a, then k lies in	(S)	(8,10)
		(T)	(9,11)
		(U)	$\left(\frac{1}{2}, \frac{1}{4}\right)$

Which of the following is the only **CORRECT** combination ?

- (A) $II \rightarrow S$ (B) $II \rightarrow Q$ (C) $IV \rightarrow U$ (D) $IV \rightarrow T$

Space for Rough Work

SEC-3 : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered.
Zero Marks : 0 In all other cases.

- Let $f(x) = |\sin x|(x-1)|(x-\pi)|(x-2\pi)|x - \frac{\pi}{2}|$ (where $\frac{9}{10} \leq x \leq 2\pi$) and $g(x) = \ln(x-1) + |x-2| + |x+3|$ (where $x > 1$). Let number of points where $f(x)$ and $g(x)$ are non differentiable are respectively λ_1, λ_2 , then $\lambda_1 + \lambda_2$ is
- If $\lim_{x \rightarrow 0} \frac{axe^x - b \log_e(1+x) - cxe^{-x}}{x^2 \sin x}$ exist and equal to 2 then value of $\left| \frac{a+c}{b} \right|$ is equal to
- The number of integral values of k for which the equation $e^x = \frac{k}{x-3}$, has exactly two solutions is -
- If $\sum_{r=1}^{44} \log_2(\sin 2r^\circ) - \sum_{r=1}^{89} \log_2(\cos r^\circ) = k$, then the value k is equal to
- Let $f(x)$ be a non constant twice derivable function defined on $\mathbb{R}$ such that $f(2+x) = f(2-x)$ and $f'\left(\frac{1}{2}\right) = 0 = f'(1)$, then minimum number of roots of the equation $f'(x) = 0$ in $(0, 4)$, is λ then $\frac{1}{\lambda}$ is equal to
- The curve $y = f(x) = ax^3 + bx^2 + cx$ is inclined at 45° to positive direction of x -axis at $(0, 0)$, but it touches x -axis at $(1, 0)$ then value of $\left(\frac{f''(1) \cdot f'''(2)}{2f(2)} \right)$ is :-

Space for Rough Work

Space for Rough Work